

[6]-Gingerol attenuates β -amyloid-induced oxidative cell death via fortifying cellular antioxidant defense system.

β -Amyloid ($A\beta$) is involved in the formation of senile plaques, the typical neuropathological marker for Alzheimer's disease (AD) and has been reported to cause apoptosis in neurons via oxidative and/or nitrosative stress. In this study, we have investigated the neuroprotective effect and molecular mechanism of [6]-gingerol, a pungent ingredient of ginger against $A\beta(25-35)$ -induced oxidative and/or nitrosative cell death in SH-SY5Y cells. [6]-Gingerol pretreatment protected against $A\beta(25-35)$ -induced cytotoxicity and apoptotic cell death such as DNA fragmentation, disruption of mitochondrial membrane potential, elevated Bax/Bcl-2 ratio, and activation of caspase-3. To elucidate the neuroprotective mechanism of [6]-gingerol, we have examined $A\beta(25-35)$ -induced oxidative and/or nitrosative stress and cellular antioxidant defense system against them. [6]-Gingerol effectively suppressed $A\beta(25-35)$ -induced intracellular accumulation of reactive oxygen and/or nitrogen species and restored $A\beta(25-35)$ -depleted endogenous antioxidant glutathione levels. Furthermore, [6]-gingerol treatment up-regulated the mRNA and protein expression of antioxidant enzymes such as γ -glutamylcysteine ligase (GCL) and heme oxygenase-1 (HO-1), the rate limiting enzymes in the glutathione biosynthesis and the degradation of heme, respectively. The expression of aforementioned antioxidant enzymes seemed to be mediated by activation of NF-E2-related factor 2 (Nrf2). These results suggest that [6]-gingerol exhibits preventive and/or therapeutic potential for the management of AD via augmentation of antioxidant capacity. Copyright © 2011 Elsevier Ltd. All rights reserved.